



Institute for Risk
and Uncertainty



Imprecise uncertainty management with uncertain numbers to facilitate trustworthy computations

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Introduction: uncertainty is nonnegligible

Scientific computations are surrounded by various forms of uncertainty, requiring appropriate treatment to maximise the credibility of computations.

Empirical information for characterization is often limited, requiring expressive uncertainty structures for credible representation, aggregation and propagation.

Bad practice: unjustified assumptions

Going overboard with assumptions of Gaussian distribution and independence without justification whenever uncertainty is involved.

Method: probability bounding approach

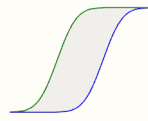
Uncertain Number unifies several uncertainty constructs including intervals, probability distributions, probability boxes (p-boxes) and Dempster-Shafer structures (DSS), plus real numbers.



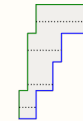
interval



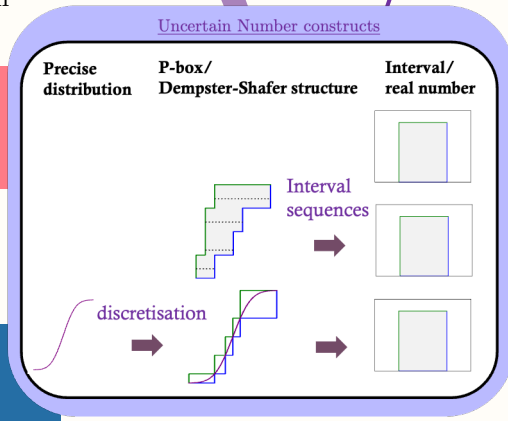
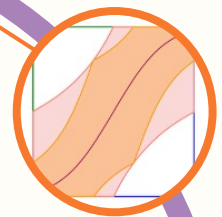
CDF



P-box



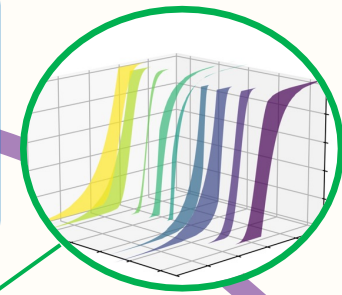
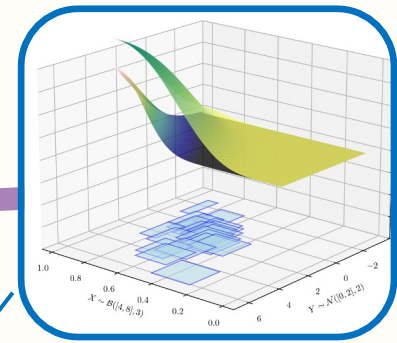
DSS



Characterisation, aggregation, and propagation

Uncertain Number enables faithful characterization and rigorous computation under mixed uncertainties and unknown dependencies, given partial empirical information:

- ♠ Qualitative or linguistic expression (e.g. 'about 7')
- ♣ Only some statistical properties are known (e.g. moments)
- ♣ Known distribution family but uncertain about parameters
- ★ Multiple sources of information, evidence, or expert elicitations
- ☞ Empirical data is scarce or imprecise



Application

- Reliability, safety and risk assessment
- Performance and design optimisation
- Uncertainty-informed decision making
- Surrogate modelling analytics

ALL YOU NEED IS

```
pip install pyuncertainnumber
```